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” PharmAIcist - AI in the Discovery of Drugs ”

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in

Artificial Intelligence & Machine Learning

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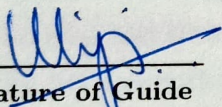
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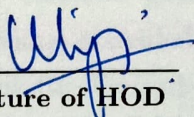
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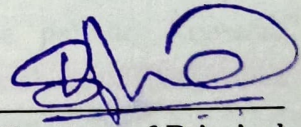


CERTIFICATE

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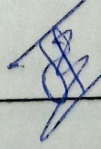
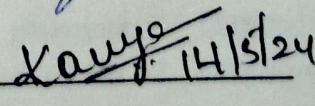

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PharmAlcist - AI in Discovery of Drugs

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Abstract

Drug discovery is the process of discovering innovative therapeutic compounds that can effectively treat diseases. Drug discovery is a complex and expensive process, taking an average of 10-15 years and costing billions of dollars. Traditional drug discovery processes are often lengthy, costly, and prone to high failure rates, largely due to the inherent complexity of biological systems and the vast chemical space of potential drug candidates. The field of drug discovery and development has greatly improved because of the use of artificial intelligence (AI) techniques. Focusing on Alzheimer's disease, a debilitating neurodegenerative disorder characterized by progressive cognitive decline, this project employs AI techniques across three components to address key challenges in drug discovery. In the first stage, bioactivity and toxicity prediction algorithms are utilized to identify potential drug candidates from the ChEMBL database, specifically targeting the protein acetylcholinesterase (AChE), which plays a crucial role in Alzheimer's pathology. Subsequently, in the second stage, a reinforcement learning architecture is employed to generate drug molecules specifically tailored for Alzheimer's disease. This phase involves a generator model employing a stacked Gated Recurrent Unit (GRU) and a predictor model trained on bioactivity data from the first stage. The generator model produces SMILES (Simplified Molecular Input Line Entry System) strings representing molecular structures, while the predictor model evaluates the bioactivity of these generated molecules. Finally, in the third stage, drug-target interaction prediction models are developed to identify potential interactions between the generated drugs and target proteins associated with Alzheimer's disease. Multiple classification models, including Dummy Classifier, Stochastic Gradient Descent Classifier, Random Forest Classifier, KNN Classifier, and Decision Tree Classifier, are trained on drug-target interaction data. The random forest predictor employed in stage one emerges as the champion model for bioactivity prediction with an impressive 86% accuracy rate, while the Extra Tree Classifier excels in toxicity prediction. In stage two, the validation is carried out using the Polypharmacology Browser (PPB2) that yields a Tanimoto similarity coefficient of 0.70, indicating significant similarity to known bioactive molecules. Again in stage three, the random forest model emerges as the most accurate, achieving a 75% accuracy rate in predicting drug-target interactions. Overall, the results of this project demonstrate the potential of AI-driven approaches in revolutionizing the drug discovery process for Alzheimer's disease.

Contents

Abstract	i
List of Figures	iv
I Introduction	1
1.1 The Problem Statement	2
1.2 Real World Application	4
1.3 Organisation of the Report	5
II Literature Survey	6
2.1 Technological Survey	6
2.2 Market Survey	8
III Problem Statement and Proposed Solution	9
3.1 Excerpt From Literature Survey	9
3.2 Problem Statement	9
3.3 Challenges	10
IV Proposed Methodology	12
4.1 Existing systems	14
4.2 Architecture	15
4.3 Training and Testing	17
V Experimentation and Results	21
5.1 Dataset Details	21

5.2	Environment setup(H/W and S/W)	25
5.2.1	Hardware Requirements:	25
5.2.2	Software Requirements:	25
5.3	Performance Analysis	26
5.4	Results	27
VI	Conclusion and Future scope	32
VII	References	34
VIII	Bibliography	37

List of Figures

4.1	Flow Diagram of proposed solution	12
4.2	System workflow for Bio-activity and Toxicity Prediction	16
4.3	System workflow for Enhanced Molecular Generation	16
4.4	System workflow for Drug-Target Interaction Prediction	17
5.1	Upload File on the Frontend	22
5.2	Graph showing the distribution of number of atoms	23
5.3	Graph showing the distribution of bond order	23
5.4	Graph showing the distribution of different types of elements	24
5.5	Illustrates the compounds within the initial, accompanied by their respective logP values.	24
5.6	Graph representing the comparision of RMSE values for different regressor models	27
5.7	Graph representing the comparision of R-squared values for different regressor models	28
5.8	Residual Plot for Random Forrest Regressor Model	28
5.9	Molecules generated through the stacked GRU along with their respective pIC50 values.	29
5.10	Graph showing the density of generated molecules in comparison to molecules in the initial dataset along with the associated pIC50 values	29
5.11	Molecules present in the Polypharmacology Browser 2 (PPB2) database which are structurally similar to the generated molecules	30
5.12	Training Performance of various classifiers employed for DTI	31
5.13	Testing Performance of various classifiers employed for DTI	31
5.14	3D structure of Acetylcholinesterase	31

Chapter I

Introduction

Drug discovery is a multifaceted process that involves the identification and development of new therapeutic compounds to address a wide range of diseases and medical conditions. It encompasses a multidisciplinary approach that integrates principles from biology, chemistry, pharmacology, and computational science. Drug discovery typically begins with target identification, where researchers identify specific molecules or biological pathways that play a crucial role in the disease process. Once a target is identified, the next step is lead compound discovery, where researchers screen large libraries of chemical compounds to identify potential candidates that can interact with the target and modulate its activity. Following lead compound identification, the optimization phase involves refining the chemical structure of the lead compounds to enhance their efficacy, selectivity, and safety profiles. This iterative process often involves medicinal chemistry techniques to modify the chemical structure of the compounds and improve their pharmacological properties. Preclinical testing involves evaluating the safety, pharmacokinetics, and efficacy of the optimized compounds in laboratory-based experiments and animal models. Finally, promising drug candidates progress to clinical testing phases, where their safety and efficacy are evaluated in human clinical trials. The drug discovery process is complex and resource-intensive, often taking many years and requiring significant investment in research and development efforts.

Traditional drug discovery methods often face significant challenges, including lengthy development timelines, high costs, and low success rates. The process of bringing a new drug to market can take over a decade and cost billions of dollars, with only a small fraction of candidate compounds successfully reaching clinical approval. This inefficiency is attributed to factors such as limited understanding of disease biology, reliance on empirical testing, and the complex nature of biological systems. Data reveals that drug discovery is a time-consuming and expensive process, taking 10-15 years and costing thousands of crores. Nine out of ten therapeutic molecules fail Phase II clinical trials and regulatory approvals. Challenges faced in drug discovery include time and cost constraints, low success rates, lack of target specificity, limited access to data, and ethical considerations. The average time for a new drug to reach the market is 12 years, and if a drug molecule does not exhibit any affinity towards the intended target, it will not be able to deliver the desired therapeutic response. Additionally, the average cost of developing a new drug and

bringing it to market is approximately 260 crore INR. The probability of a drug entering clinical trials to eventually receive approval is about 12

Artificial intelligence (AI) has emerged as a promising tool in drug discovery, offering solutions to overcome many of the challenges associated with traditional methods. AI techniques, including machine learning and deep learning algorithms, enable the analysis of large datasets, prediction of molecular properties, and identification of novel drug candidates with greater speed and accuracy. By leveraging AI, researchers can expedite the drug discovery process, reduce costs, and improve the likelihood of success.

Alzheimer's disease (AD) is a progressive neurodegenerative disorder characterized by memory loss, cognitive decline, and behavioral changes, stemming from the accumulation of abnormal protein deposits in the brain, such as beta-amyloid plaques and tau tangles. [20] As the disease advances, individuals may struggle with daily tasks, communication, and recognizing loved ones. With no cure currently available, AD poses significant challenges for patients and healthcare systems worldwide, leading to a growing demand for effective treatments. A crucial aspect of AD involves the disruption of neurotransmitters, particularly acetylcholine, which plays a crucial role in memory and learning. The reduction in acetylcholine levels due to the damage of cholinergic neurons is a target for drug discovery. While acetylcholinesterase inhibitors are commonly used to manage AD symptoms by temporarily elevating acetylcholine levels, they do not address the disease's underlying progression. Consequently, there's a pressing need for innovative therapeutic strategies that target the fundamental mechanisms of AD pathology. In this context, the integration of artificial intelligence (AI) in drug discovery emerges as a promising avenue, offering opportunities to accelerate the development of novel treatments for AD.

In this implementation, the aim is to apply AI techniques to drug discovery for Alzheimer's disease, with a focus on targeting the acetylcholinesterase enzyme. The project comprises three components: (1) prediction of bioactivity and toxicity of molecules using machine learning algorithms, (2) generation of potential drug candidates using AI-driven molecular design methods, and (3) prediction of drug-target interactions using computational models trained on large datasets. By integrating AI into each stage of the drug discovery process, the goal is to accelerate the identification of effective treatments for Alzheimer's disease while minimizing costs and resource requirements.

1.1 The Problem Statement

In drug discovery, researchers face numerous challenges that impede the development of new therapeutics. One of the primary issues is the vast amount of data generated from various sources, including genomic data, chemical databases, and medical literature. This data overload makes

it difficult for researchers to identify meaningful patterns and relationships that could lead to the discovery of effective drugs. Additionally, the process of target identification and validation is time-consuming and resource-intensive, often requiring years of experimentation to confirm a target's role in disease progression. The key problems faced in traditional drug discovery methods are mentioned below:

- **Data Overload:** Drug discovery involves analyzing vast amounts of data from multiple sources, including genomic data, chemical databases, and scientific literature. However, the sheer volume of information often overwhelms researchers, making it challenging to identify relevant insights and patterns.
- **Target Identification and Validation:** Identifying and validating potential drug targets is a crucial step in the drug discovery process. However, this process is time-consuming and resource-intensive, requiring extensive experimentation to confirm a target's role in disease progression.
- **Trial and Error in Drug Design:** Traditional drug design methods often rely on trial and error, resulting in a lengthy and costly process. Researchers struggle to design molecules with the desired properties, such as potency and safety, leading to a high rate of failure in preclinical and clinical trials.
- **Prediction of Drug Efficacy and Safety:** Predicting the efficacy and safety of drug candidates is challenging, as it relies on extrapolating data from animal studies and small-scale human trials. However, these predictions may not accurately reflect how the drug will perform in larger populations, leading to uncertainties in the drug development process.
- **Cost and Time Constraints:** Bringing a new drug to market is a lengthy and expensive process, often taking upwards of 15 years and costing millions of dollars. The high costs and long timelines associated with drug development contribute to the overall inefficiency of the process.
- **High Failure Rates:** Despite significant investment in drug discovery research, the majority of drug candidates fail to gain regulatory approval. This high failure rate not only delays the availability of new treatments but also represents a significant loss of resources for pharmaceutical companies and research institutions.

In conclusion, the challenges inherent in the traditional drug discovery process underscore the critical importance of adopting innovative approaches to overcome these obstacles. By leveraging

cutting-edge technologies and collaborative efforts, researchers can streamline drug development processes and expedite the delivery of safe and effective therapies to patients in need. Moreover, ongoing advancements in artificial intelligence and computational techniques hold tremendous promise for revolutionizing drug discovery and driving transformative breakthroughs in medical science.

- **Virtual Screening:** AI algorithms can analyze vast databases of chemical compounds and predict their potential as drug candidates based on their structure and properties. This significantly expedites the initial screening process, allowing researchers to prioritize compounds with the highest likelihood of success.
- **Drug Repurposing:** AI can identify existing drugs that may be effective in treating different diseases than originally intended. By analyzing large datasets of drug interactions, molecular structures, and biological pathways, AI algorithms can pinpoint potential candidates for repurposing, saving time and resources in drug development.
- **Bioactivity and Toxicity Prediction:** AI algorithms analyze molecular structures and biological data to predict the bioactivity and toxicity of drug candidates, aiding in identifying compounds with therapeutic effects and minimizing adverse reactions. Leveraging machine learning and deep learning techniques, these models learn from vast datasets to accurately predict compound interactions, streamlining drug discovery.
- **Generation of Molecules:** AI-based generative models employ techniques like reinforcement learning and variational autoencoders to produce novel molecular structures with desired properties, accelerating the search for lead compounds. By optimizing parameters such as potency and specificity, these models explore vast chemical spaces and generate diverse sets of molecules for drug design.
- **Drug Target Interactions:** AI facilitates the prediction and analysis of interactions between drugs and their molecular targets, providing insights into mechanisms of action and potential side effects. Through computational methods and experimental data integration, AI predicts drug-target binding affinities and optimizes drug design for enhanced efficacy and safety.

1.2 Real World Application

Artificial Intelligence (AI) in drug discovery represents a groundbreaking frontier in pharmaceutical research, promising to revolutionize the process of drug development and innovation. By

harnessing the computational power of AI algorithms and machine learning models, researchers are empowered to expedite the identification of novel therapeutic candidates and optimize drug design with unprecedented efficiency and accuracy.

1.3 Organisation of the Report

The project report is organized into several chapters to systematically present the research findings and methodology. Chapter 2 comprises a comprehensive literature survey, encompassing both technological and market surveys to provide a thorough understanding of the existing landscape in drug discovery. Chapter 3 outlines the problem statement, proposed solution, and includes excerpts from the literature survey, emphasizing the motivation and challenges addressed by the project. In Chapter 4, the proposed methodology is detailed, along with an examination of existing systems and the architecture of the solution, including training and testing methods. Chapter 5 focuses on experimentation and results, covering dataset details, environment setup (hardware and software), testing procedures, performance analysis, and the presentation of results. Finally, Chapter 6 concludes the report, summarizing the key findings, highlighting contributions to the field, and suggesting avenues for future research and development.

Chapter II

Literature Survey

In this section, an exploration of the current landscape of technologies and methodologies relevant to the research objectives is undertaken. This section offers a comprehensive overview of existing tools, techniques, and advancements in the field of artificial intelligence, drug discovery, and molecular modeling. By examining the latest developments and trends, the aim is to identify key areas of innovation and leverage cutting-edge technologies to propel the research forward.

2.1 Technological Survey

Ross et. al [1] wrote a paper on comparing different artificial intelligence (AI) methods for modeling pharmaceutical Quantitative Structure-Activity Relationships (QSARs). The authors conducted a comparative analysis of various AI techniques, including neural networks, decision trees, and support vector machines, to determine their effectiveness in predicting the activity of pharmaceutical compounds based on their chemical structures. The study was carried out by the Biomolecular Modelling Laboratory at the Imperial Cancer Research Fund in London, UK. The findings of the paper contribute to understanding the performance and suitability of different AI methods for modeling QSARs in the pharmaceutical industry.

Hamza et al [2] introduced a paper focused on the use of Convolutional Neural Networks (CNNs) for bioactivity prediction in the field of drug discovery. The authors explore the application of CNNs in analyzing molecular structures and predicting their bioactivity against specific targets. They discuss the advantages of CNNs in capturing spatial features and patterns in chemical compounds, which are essential for accurate bioactivity prediction. The findings demonstrate the effectiveness of CNNs in accurately predicting bioactivity, highlighting their potential for enhancing drug discovery efforts.

Mayr et al. [3] present DeepTox, a deep learning model designed for accurately predicting compound toxicity in drug discovery. By incorporating convolutional and recurrent neural network layers, DeepTox captures both spatial and temporal features of chemical compounds. The model is trained on a large dataset of compounds with known toxicity labels and achieves high accuracy in toxicity prediction, outperforming traditional machine learning methods. DeepTox offers scalability and the potential to handle complex datasets, demonstrating its potential to enhance

drug safety and efficacy by identifying compounds with low toxicity.

Yan-Bin Wang and his team [4] introduce a novel approach for predicting drug target interactions using deep learning techniques, specifically the Long Short-Term Memory (LSTM) neural network. The authors address the challenge of accurately predicting the interactions between drugs and their target proteins, which is crucial in drug discovery. By leveraging the temporal dependencies and sequential nature of drug-target interaction data, the LSTM network is trained to learn the underlying patterns and relationships between drugs and targets.

Mariya et.al [5] introduce ReLeaSE (Reinforcement Learning for Structural Evolution), a groundbreaking computational strategy rooted in deep and reinforcement learning for the de novo design of molecules with predefined properties. ReLeaSE integrates two distinct deep neural networks—generative and predictive—trained separately and collaboratively used to generate targeted chemical libraries. Utilizing simplified molecular-input line-entry system (SMILES) strings, the generative models employ a stack-augmented memory network to produce feasible chemical structures, while predictive models forecast desired properties.

Marcus et al. [6] introduce an innovative approach for refining a sequence-Based generative model in order to design molecules from the ground up, utilizing augmented episodic likelihood for generating structures with predefined properties. The model showcases versatility in tasks like generating sulfur-free molecules, Celecoxib analogues for scaffold hopping, and compounds predicted to be active against the dopamine receptor type 2 (DRD2). Compared to traditional policy gradient methods, the model exhibits remarkable performance in reflecting the underlying probability distribution of the Prior. The capacity to explore chemical space outside the Prior and generate structures with over 95% predicted activity against DRD2 highlights its potential in the discovery of drugs. The study addresses considerations such as Prior qualities, hyperparameters' robustness, and the necessity for multi-parametric scoring functions in the projects involving the discovery of drugs, emphasizing the promise of sequence-Based generative models for molecular design.

Qichang Zhao et al. [7] proposed an innovative bio-inspired model named HyperAttentionDTI, leveraging convolutional neural networks (CNNs) and the attention mechanism for predicting drug-target interactions (DTIs). Unlike existing methods, HyperAttentionDTI aims to model complex non-covalent inter-molecular interactions among atoms and amino acids. The model employs deep CNNs to learn feature matrices of drugs and proteins, while the attention mechanism is utilized to assign an attention vector to each atom or amino acid, capturing intricate interactions. Evaluation on three benchmark datasets demonstrated that HyperAttentionDTI outperformed state-of-the-art baselines, indicating its efficacy in DTI prediction. Furthermore, a case study on

human Gamma-aminobutyric acid receptors highlighted the model's potential as a powerful tool for predicting DTIs.

2.2 Market Survey

The drug discovery market is characterized by significant costs and complexities, with the global pharmaceutical industry investing billions of dollars annually. According to industry reports, the average cost of developing a new drug and bringing it to market exceeds \$2.5 billion globally, a figure that continues to rise due to increasing research and development expenses, stringent regulatory requirements, and the high failure rates associated with drug development. Additionally, the process of bringing a new drug to market typically takes 10-15 years, further adding to the overall cost and time investment. Despite these challenges, the drug discovery market remains robust, driven by rising demand for innovative treatments, advances in biotechnology and genomics, and growing prevalence of chronic diseases worldwide.

Based on current trends, the average cost of developing and bringing a new drug to the Indian market stands at approximately 260 crore INR. However, the journey from drug discovery to market approval is fraught with challenges, as only around 12% of drug candidates entering clinical trials eventually receive approval. Moreover, a staggering 75% of drug candidates fail in clinical trials, primarily due to efficacy or safety concerns. These statistics underscore the immense financial risks and uncertainties associated with drug development, highlighting the critical need for robust research and development strategies to improve the success rate and cost-effectiveness of bringing new drugs to market.

Numerous organizations are actively engaged in drug discovery research, ranging from large pharmaceutical companies to academic institutions and government agencies. Leading pharmaceutical companies such as Pfizer, Novartis, and Roche invest heavily in research and development, allocating significant resources to drug discovery initiatives. Academic research institutions, including universities and medical centers, play a crucial role in advancing drug discovery through fundamental research and collaboration with industry partners. Additionally, government agencies such as the National Institutes of Health (NIH) and the European Medicines Agency (EMA) provide funding and support for drug discovery research, further fueling innovation in the field.

Chapter III

Problem Statement and Proposed Solution

3.1 Excerpt From Literature Survey

The literature survey encompasses a variety of AI-based methodologies applied to pharmaceutical research, spanning from traditional machine learning techniques to advanced deep learning models. King et al. conducted a comparative analysis of AI methods for modeling pharmaceutical Quantitative Structure-Activity Relationships (QSARs), shedding light on the effectiveness of neural networks, decision trees, and support vector machines. Hensabali et al. explored the utilization of Convolutional Neural Networks (CNNs) for bioactivity prediction in drug discovery, emphasizing their ability to capture spatial features in molecular structures. Additionally, Wang et al. introduced a novel approach employing Long Short-Term Memory (LSTM) neural networks for predicting drug-target interactions, addressing the challenge of accurately identifying interactions crucial in drug discovery. Furthermore, Olivecrona et al. and Zhao et al. introduced ReLeaSE, a strategy for de novo molecule design, and HyperAttentionDTI, a model for predicting drug-protein interactions, respectively, showcasing the potential of deep learning in revolutionizing drug discovery and molecular design. These studies collectively underscore the growing significance of AI in pharmaceutical research and its potential to advance drug discovery and development processes.

3.2 Problem Statement

Traditional drug discovery methods face significant challenges that hinder the timely and cost-effective development of new therapeutics. One of the primary obstacles is the prolonged development timeline, which typically spans over a decade from initial discovery to market approval. This protracted process incurs exorbitant costs, often amounting to billions of dollars, without guaranteeing success. Moreover, the reliance on empirical testing and trial and error in drug design contributes to the high failure rates observed in preclinical and clinical trials. Despite substantial investments in research and development, the vast majority of candidate compounds

fail to progress beyond these components, leading to significant resource wastage and delays in bringing new treatments to patients. Additionally, accurately predicting the efficacy and safety of drug candidates remains a formidable challenge, resulting in uncertainties that further prolong the drug development timeline. These challenges underscore the pressing need for innovative approaches to drug discovery that can streamline the process, enhance success rates, and expedite the delivery of safe and effective therapies to patients in need.

3.3 Challenges

The challenges encountered in applying artificial intelligence (AI) to drug discovery are explored in this section and are as follows:

- **Data Complexity:** One of the primary challenges in AI-driven drug discovery is the complexity and heterogeneity of biomedical data. Biological systems are intricate and multifaceted, encompassing diverse molecular interactions, pathways, and phenotypic responses. Integrating and interpreting data from various sources, such as genomics, proteomics, and metabolomics, presents significant computational and analytical challenges.
- **Data Quality and Availability:** Another key challenge is the quality and availability of data, particularly labeled datasets for training machine learning models. Biomedical data are often sparse, noisy, and incomplete, leading to difficulties in model generalization and prediction accuracy. Additionally, access to comprehensive and well-curated datasets is limited, hindering the development and validation of AI algorithms for drug discovery.
- **Interpretability and Explainability:** The inherent complexity of AI models poses challenges in interpreting and explaining their predictions and decisions, especially in the context of drug discovery. Understanding the underlying biological mechanisms and features driving model predictions is crucial for gaining insights into drug-target interactions and identifying potential drug candidates. Enhancing the interpretability and explainability of AI models is essential for building trust and confidence in their use in drug discovery.
- **Computational Resources:** AI-based drug discovery methods often require substantial computational resources, including high-performance computing infrastructure and scalable algorithms. Processing and analyzing large-scale biological datasets, conducting virtual screening, and optimizing molecular structures demand significant computational power and storage capacity. Access to adequate computational resources presents a barrier for researchers and institutions with limited infrastructure.

- **Regulatory and Ethical Considerations:** Integrating AI technologies into drug discovery workflows raises regulatory and ethical considerations related to patient safety, data privacy, and intellectual property rights. Ensuring compliance with regulatory standards and guidelines, such as those set forth by regulatory agencies like the FDA and EMA, is essential for the successful translation of AI-driven discoveries into clinical applications. Addressing ethical concerns surrounding data usage, transparency, and fairness in AI algorithms is also paramount for fostering public trust and acceptance.

Addressing these challenges of AI in drug discovery necessitates collaborative efforts from researchers, industry stakeholders, and regulatory bodies.

Chapter IV

Proposed Methodology

The proposed methodology integrates artificial intelligence (AI) techniques into the drug discovery process, with a focus on addressing the complexities of Alzheimer's disease. Leveraging AI algorithms and computational methods, this project aims to expedite the identification of potential drug candidates for Alzheimer's disease and enhance the efficiency of the drug discovery pipeline. The solution comprises three components, each designed to tackle specific challenges in drug discovery and optimization, ultimately contributing to the development of novel therapies for Alzheimer's disease.

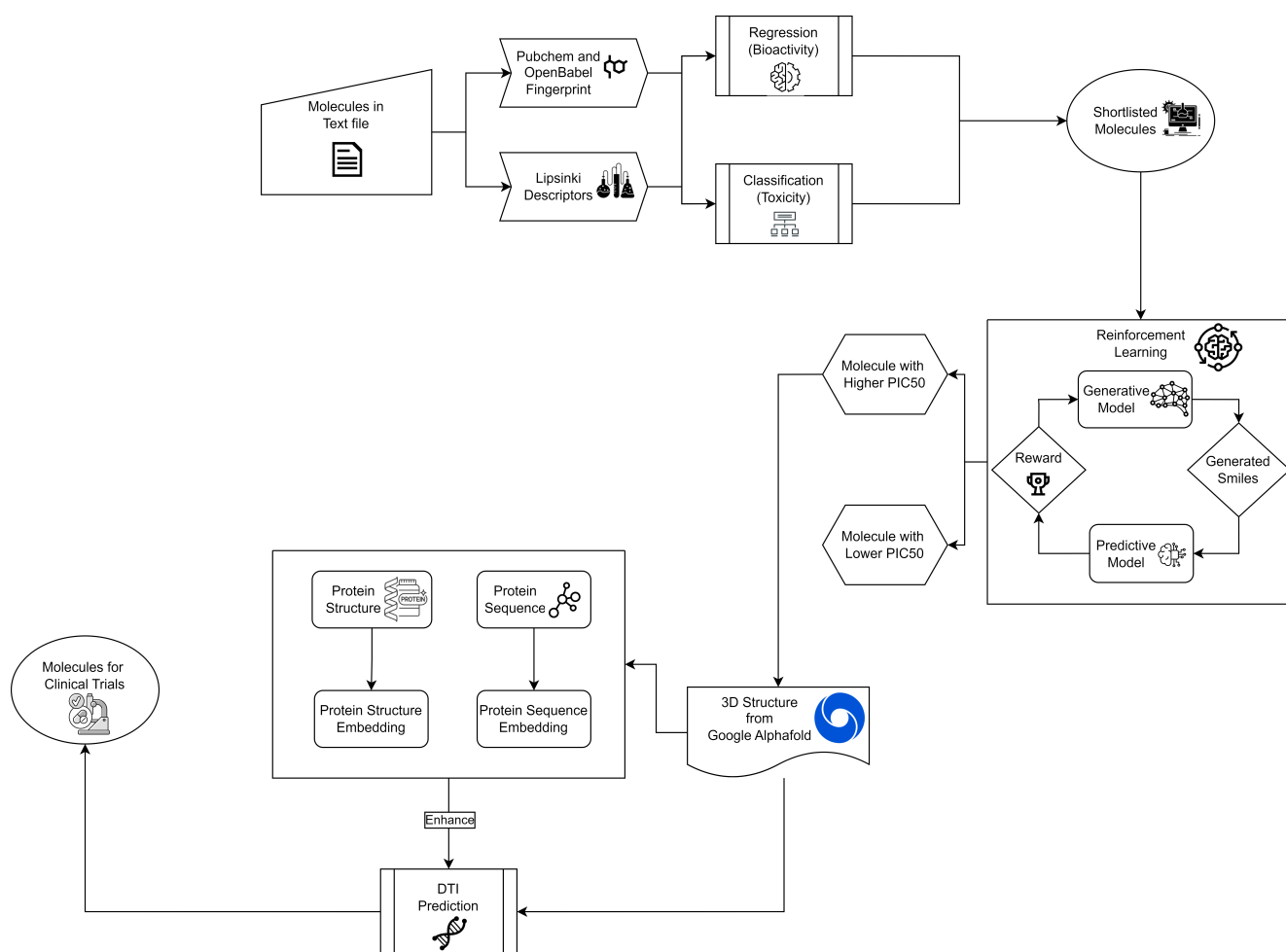


Figure 4.1: Flow Diagram of proposed solution

The three components employed are as follows:

1. **Bioactivity and Toxicity Prediction:** To predict the bioactivity and toxicity of molecules targeting Acetylcholinesterase, which is a key strategy in combating Alzheimer's disease, the project employs machine learning (ML) models. These models are designed to analyze chemical compounds and predict their pharmacological properties, specifically focusing on their interaction with the AChE protein. By leveraging ML techniques, the project aims to identify potential therapeutic compounds that can effectively inhibit AChE activity, thereby mitigating the progression of Alzheimer's disease. This approach involves training ML models on large datasets of chemical compounds with known bioactivity and toxicity profiles, enabling the accurate prediction of the pharmacological properties of novel compounds. The ultimate goal is to facilitate the discovery of new drug candidates with the potential to treat Alzheimer's disease by targeting the AChE protein.
2. **Enhanced Molecular Generation:** Enhanced molecular generation builds on the results obtained by prediction of bioactivity and toxicity component where promising compounds were identified for further exploration in drug discovery. Leveraging the wealth of information obtained from bioactivity and toxicity predictions, the primary goal of this component is to leverage the bioactivity and toxicity data to discover novel drugs aimed at preventing Alzheimer's disease. The aim of this component is to enhance the drug discovery process using advanced machine learning and deep learning techniques, thereby reducing traditional time and resource requirements. Through systematic screening and computational modeling, the goal is to expedite the identification of lead compounds with optimal therapeutic efficacy and safety profiles, addressing the urgent need for Alzheimer's disease treatments.
3. **Drug-Target Interactions:** The objective of this component, building upon the findings from the previous component, is to utilize machine learning techniques for examining the structural relationships between drugs and target proteins, specifically focusing on acetylcholinesterase. The aim is to predict drug-target interactions, furthering drug discovery efforts by uncovering key structural features governing their binding affinity and efficacy. Through thorough analysis and modeling of these interactions, the goal is to expedite the drug discovery process. This analysis aids in determining whether the relationship between drugs and proteins is active or inactive, providing valuable insights into the efficacy and potential of various compounds in targeting specific proteins.

Together, these three components form a comprehensive AI-driven approach to drug discovery for Alzheimer's disease, encompassing the prediction of bioactivity and toxicity, the generation of novel drug molecules, and the elucidation of drug-target interactions. By leveraging the power of

AI and computational methods, this solution aims to accelerate the discovery and development of effective treatments for Alzheimer's disease.

4.1 Existing systems

Traditional drug discovery processes have long relied on established systems and methodologies to identify, develop, and evaluate potential therapeutic compounds. These systems, rooted in empirical experimentation and scientific principles, form the foundation of pharmaceutical research and development. In this section, we delve into the existing systems that have historically shaped the landscape of drug discovery, highlighting their significance and contributions to the field.

1. High-Throughput Screening (HTS):

- HTS involves the rapid screening of large compound libraries to identify molecules with desired biological activity.
- Automated robotic systems enable the screening of thousands to millions of compounds against biological targets, facilitating the identification of potential drug candidates.
- Disadvantage: Despite its efficiency in screening large compound libraries, HTS often generates a high rate of false positives and requires subsequent validation through secondary assays.

2. Structure-Based Drug Design (SBDD):

- SBDD relies on the knowledge of a target's three-dimensional structure to design molecules that interact with specific binding sites.
- Computational modeling techniques, such as molecular docking and virtual screening, play a crucial role in SBDD by predicting the binding affinity and interactions between drug candidates and target proteins.
- Disadvantage: SBDD heavily relies on accurate structural information, which may not be available for all target proteins. Additionally, predicting the binding affinity and selectivity of compounds using computational methods can be challenging.

3. Phenotypic Screening:

- Phenotypic screening evaluates the biological activity of compounds based on observable phenotypic changes in cells or organisms.
- Unlike target-based approaches, phenotypic screening does not require prior knowledge of the target or mechanism of action, making it suitable for identifying novel drug candidates with unknown modes of action.

- Disadvantage: Phenotypic screening often lacks mechanistic insights into the mode of action of identified compounds, leading to challenges in target identification and optimization.

4. Fragment-Based Drug Design (FBDD):

- FBDD involves the identification of small, fragment-sized molecules that bind to a target protein, which are then elaborated into larger, more potent compounds.
- By focusing on small, structurally diverse fragments, FBDD enables the exploration of chemical space more efficiently and facilitates the identification of high-quality lead compounds.
- Disadvantage: FBDD requires extensive chemical synthesis and optimization efforts to transform fragment hits into lead compounds, which can be time-consuming and resource-intensive. Additionally, fragment-based libraries may not cover the full chemical diversity required for effective lead identification.

These existing systems of traditional drug discovery processes have played a pivotal role in advancing pharmaceutical research and drug development. While each system has its strengths and limitations, their integration and iterative refinement continue to drive innovation in the quest for new therapeutics. As we explore emerging technologies and methodologies, it is essential to recognize and build upon the foundations established by these systems to further accelerate drug discovery efforts and address unmet medical needs.

4.2 Architecture

The system workflow for each of the three components are explained below. Each component's architecture provides information on data preprocessing, model training, and validation stages, offering insights into the computational framework underpinning the drug discovery process.

Prediction of bioactivity and toxicity of molecules:

In the first stage, AI algorithms are employed to predict the bioactivity and toxicity of molecules sourced from the ChEMBL database, specifically targeting the acetylcholinesterase enzyme associated with Alzheimer's disease. Using machine learning models such as Random Forest and Extra Tree Classifier, the bioactivity and toxicity profiles of candidate molecules are evaluated, allowing for the identification of potential drug candidates with favorable pharmacological properties.

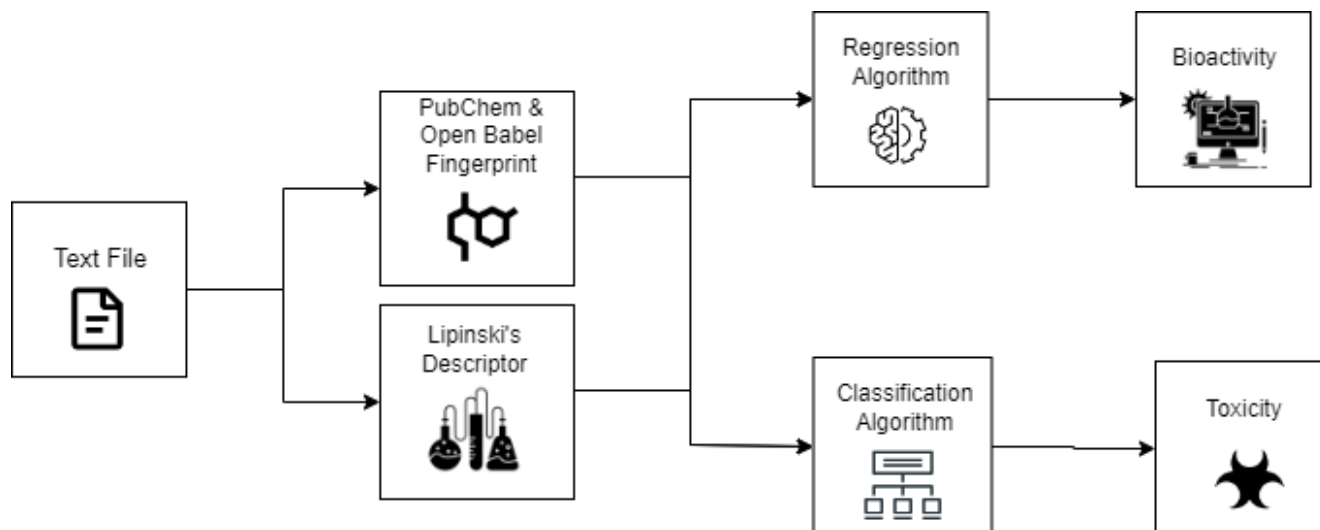


Figure 4.2: System workflow for Bio-activity and Toxicity Prediction

Enhanced Molecular Generation:

The second stage focuses on generating novel drug-like molecules using AI-driven generative models. By employing techniques such as stacked Gated Recurrent Unit (GRU) architecture and reinforcement learning, the project aims to generate diverse molecular structures optimized for targeting Alzheimer's disease. The generated molecules are assessed for their similarity to known drug compounds and their potential therapeutic efficacy, providing valuable insights for further optimization.

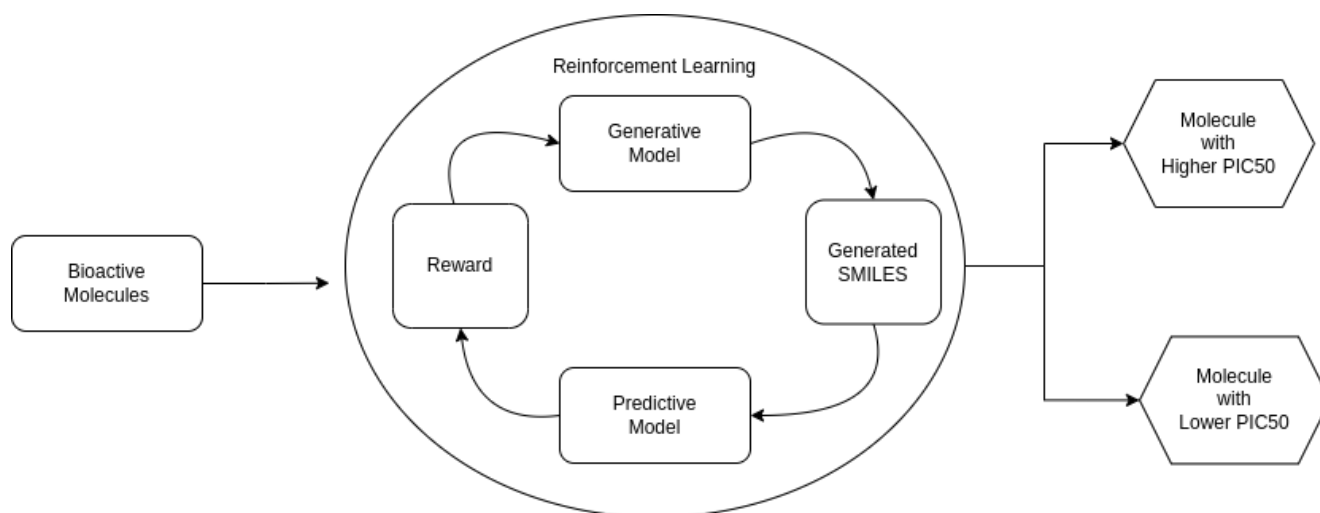


Figure 4.3: System workflow for Enhanced Molecular Generation

Prediction of Drug-Target Interactions:

In the final stage, AI algorithms are utilized to predict the interactions between generated drug molecules and their target proteins, including acetylcholinesterase and other relevant biomolecules associated with Alzheimer's disease. By training machine learning classifiers on a dataset of drug-target interactions, the project seeks to elucidate the mechanisms of action of candidate

molecules and prioritize lead compounds for preclinical testing. Additionally, the integration of protein structure embeddings obtained from advanced computational methods such as AlphaFold enhances the accuracy and robustness of the drug-target interaction predictions.

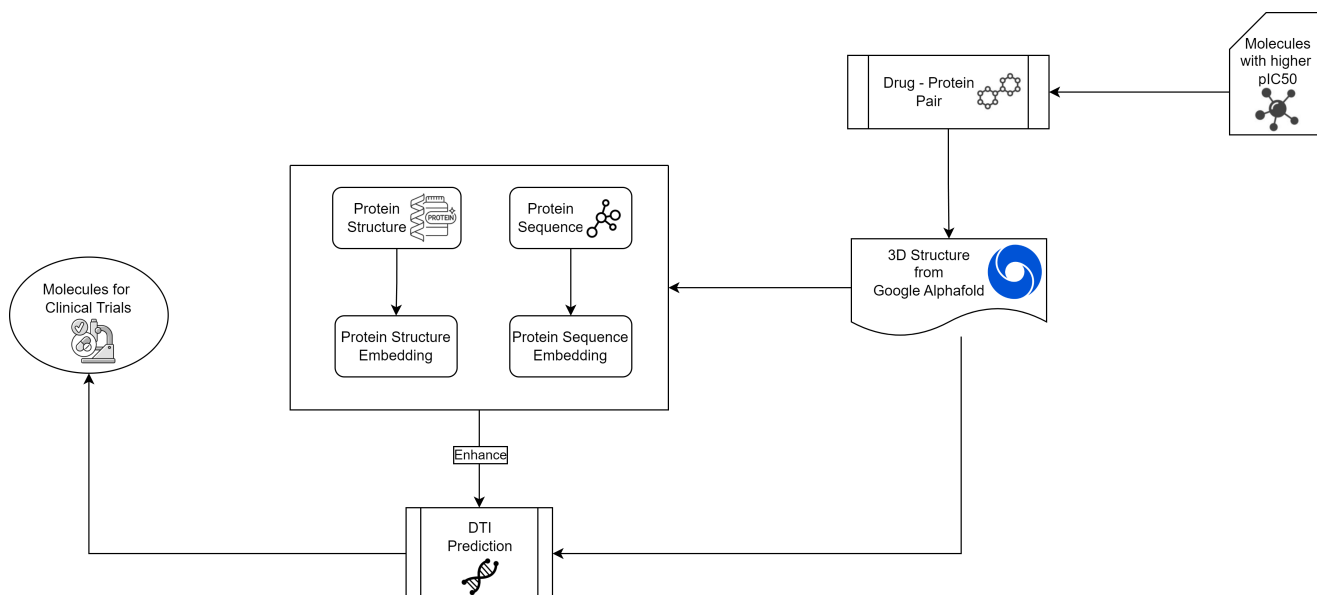


Figure 4.4: System workflow for Drug-Target Interaction Prediction

4.3 Training and Testing

This section delves into the training and evaluating processes of various models across all three components of the project: bioactivity and toxicity prediction, enhanced molecular generation, and drug-target interaction. Through meticulous experimentation, the efficacy and performance of different machine learning and deep learning algorithms are assessed, contributing to informed model selection in the drug discovery pipeline.

Prediction of Drug-Target Interactions:

The methodology begins with the acquisition of data from the ChEMBL database [10], in SMILES format [11], a compact representation of molecular structures. Upon obtaining the SMILES strings, preprocessing is conducted to ensure data integrity and consistency. The SMILES strings are then converted into PubChem fingerprints [12] using the PaDEL Descriptor tool [13]. These fingerprints serve as local features, capturing specific molecular characteristics such as substructures and molecular weight. In parallel, the toxicity of chemical compounds is predicted using a similar approach. Compounds in SMILES format from the datasets are converted into fingerprints using Open Babel [14], a chemical toolbox designed for molecular modeling. Additionally, global features are extracted using the Lipsinki Descriptor [15], encompassing ADME properties which stand for Absorption, Distribution, Metabolism, and Excretion properties. These properties refer to the processes that a drug undergoes within the body, including how it is absorbed into the

bloodstream, distributed to various tissues, metabolized by enzymes, and eventually eliminated from the body. This comprehensive feature extraction process provides a holistic representation of the chemical compounds.

With the extracted features in hand, various regression algorithms are employed to predict the bioactivity of the compounds. These algorithms include Decision Tree Regressor, Linear Regression, Ridge Regression, ExtraTreesRegressor, Bagging Regressor, XGB Regressor, AdaBoost Regressor, KNeighbors Regressor, ExtraTreeRegressor, and Random Forest Regressor. The models are trained on known bioactivity data, where the relationship between compound features and their corresponding PIC50 values is learned. The PIC50 value [16], also known as the negative logarithm of the IC50 concentration, is a measure of the potency of a compound in inhibiting a biological process by 50%. In drug discovery, predicting the PIC50 values of chemical compounds provides valuable insights into their potential efficacy. The trained models are then utilized to predict the PIC50 values of unseen compounds, providing insights into their potential efficacy as therapeutic agents.

For toxicity prediction, the SMILES strings are utilized once again, converted into fingerprints by Open Babel. These fingerprints serve as input features for the extra tree algorithm, a machine learning algorithm suitable for classification tasks. The extra tree algorithm is trained on toxicity data, enabling it to classify compounds based on their toxicity levels. This predictive model aids in assessing the safety profile of potential drug candidates, ensuring that only compounds with acceptable toxicity profiles are further considered for development.

Enhanced Molecular Generation:

The methodology for molecular prediction in this study begins with obtaining bioactivity and toxicity data from the prediction components of bioactivity and toxicity. These datasets serve as the foundation for the subsequent phases of the process. Leveraging the bioactivity data, a Random Forest predictor is employed to forecast PIC50 values, representing compound potency. This predictor is selected for its efficacy in forecasting bioactivity, outperforming other regression models due to its robustness in handling the complexities of Acetylcholinesterase-related compounds, crucial in Alzheimer's disease research. The input compounds, represented in SMILES format, undergo adept processing by the Random Forest model, with their conversion into PubChem fingerprints facilitating the subsequent feature selection process. This meticulous feature selection step refines the dataset to include only essential features essential for predictive modeling, thereby enhancing the efficiency of the predictive model.

Concurrently, a stacked Gated Recurrent Unit (GRU) architecture is deployed as a generative model to produce SMILES strings representing molecular structures. Configured with hyperpa-

rameters conducive to diversity and fidelity in structure generation, the GRU model is trained to minimize cross-entropy loss between predicted and ground truth symbols, ensuring the accuracy of the generated SMILES strings. The GRU generator exhibits versatility beyond SMILES generation, extending its functionality to predict compound potency (pIC50) and incorporating a filtering mechanism to exclude invalid molecular structures, thereby ensuring the generation of chemically meaningful compounds.

Integration of reinforcement learning (RL) for optimization marks a pivotal phase in the methodology, where the GRU generator and the Random Forest predictor synergistically unite in a cohesive pipeline. The policy gradient algorithm guides iterative updates of the generator, orchestrated by a well-defined reward function encapsulating the strategy. By incentivizing the production of molecular structures aligned with desired logP values [19] and pharmacologically relevant properties, the RL integration optimizes molecular properties, fostering a refined exploration of chemical space. Through this integrated approach, the methodology aims to strike an optimal balance between quantitative logP prediction and the generation of molecular structures with desirable properties for Alzheimer’s disease research and drug discovery.

The reward function is provided in Eq.1

$$R(s) = \exp\left(\frac{\text{predictor}(s)}{3}\right) \quad (1)$$

The policy gradient loss is provided in Eq. 2

$$L(S | \theta) = -\frac{1}{n} \sum_{i=1}^{|S|} R_i \sum_{j=1}^{\text{length}(s_i)} R_i \cdot \gamma^i \cdot \log p(s_i | s_{i-1} \theta) \quad (2)$$

Here, R_i is the reward obtained at time step i , γ is the discount factor, and $p(s_i | s_0 \dots s_{i-1}, \theta)$ is the probability of the next character given the prefix, obtained from the generator.

The comprehensive methodology successfully generated SMILES formatted molecular drugs with elevated pIC50 values, indicating their potential as highly bioactive compounds. The pIC50 values, spanning from 0 to 14, served as a quantitative measure of bioactivity, with higher values indicating increased potency. The research’s holistic approach, integrating predictive modeling, generative techniques, and reinforcement learning, holds significance in advancing computational drug discovery, particularly in the realm of Alzheimer’s disease.

Prediction of Drug-Target Interactions:

The drug-target interaction methodology involves several steps aimed at predicting interactions between drugs and target proteins. Initially, data is sourced from the ChEMBL database for drug information and the UniProt website [17] for protein information. SMILES strings for drugs are converted into PubChem fingerprints, capturing their structural features. Concurrently, pro-

tein sequences are extracted and processed to obtain protein sequence embeddings and structure embeddings using Google DeepMind AlphaFold [18], enriching the dataset with additional information.

The dataset, consisting of drug features (PubChem fingerprints), protein features, and drug-target interaction scores, is prepared for model training. Multiple classifier models, including Dummy Classifier, Stochastic Gradient Descent Classifier, Random Forest Classifier, KNN Classifier, and Decision Tree Classifier, are selected as baseline models due to their promising performance in preliminary evaluations.

These baseline models are trained on the prepared dataset to predict drug-target interactions. During training, the models learn the relationships between drug and protein features and their corresponding interaction scores. Performance metrics such as accuracy, precision, recall, and F1 score are used to evaluate the models' performance on validation data. Furthermore, to enhance model performance, additional features derived from protein structures and sequences are incorporated into the dataset. The protein sequence embeddings and structure embeddings obtained from AlphaFold provide deeper insights into the structural characteristics of target proteins. These embeddings are included as features alongside drug fingerprints, enriching the dataset and enabling the models to better capture the nuances of drug-target interactions.

Chapter V

Experimentation and Results

The experimentation and results section serves as a comprehensive exploration of the methodologies employed, the environments utilized, the performance metrics analyzed, and the outcomes obtained throughout the project. It delves into the specifics of the datasets utilized, providing insights into their composition, size, and relevance to the study. Additionally, the section sheds light on the setup of the computational environments, including both hardware and software configurations, which played a crucial role in executing the experiments effectively. Furthermore, the performance analysis offers a detailed examination of the models’ predictive capabilities, highlighting metrics such as accuracy, precision, recall. Finally, the results segment encapsulates the findings and discoveries unearthed during the project, providing a comprehensive overview of the outcomes achieved and their implications for the field of drug discovery.

5.1 Dataset Details

Prediction of bioactivity and toxicity of molecules

The project relies on two robust data sources to facilitate accurate prediction of bioactivity and toxicity of chemical compounds. The first source is the ChEMBL database, a widely recognized and extensively utilized resource in drug discovery research. This database encompasses a vast array of bioactivity data, including information on small molecules and their interactions with protein targets. With its inclusion of data from scientific literature, patents, and various databases, ChEMBL offers researchers a comprehensive collection of experimentally measured bioactivity information. It comprises over 15 million data points, covering approximately 1.9 million compounds and their interactions with nearly 8000 protein targets. The research of structure-activity correlations (SAR) and the discovery of new drug candidates are all made possible by the vast amounts of data that are available. Moreover, ChEMBL’s accessibility to the academic and non-profit community enhances its utility in bioactivity prediction and drug discovery research. Along with the ChEMBL database, data from the Tox21 and TOXNET datasets was incorporated to accurately assess the toxicity of chemical compounds. The Tox21 dataset is the result of coopera-

tion between numerous American scientific institutions and governmental organizations, focusing on "Toxicology in the 21st Century." It includes a wide range of chemical substances and a thorough set of experimental findings that describe their toxicity profiles in great detail. However, the TOXNET dataset, which is managed by the National Library of Medicine, is a useful resource for information on toxicology, dangerous substances, and environmental health. It encompasses diverse data sources, including databases such as HSDB (Hazardous Substances Data Bank) and CCRIS (Chemical Carcinogenesis Research Information System). By leveraging the rich and diverse data available in these datasets, we were able to establish a robust foundation for accurately detecting and evaluating the toxicity of chemical compounds. The utilization of the ChEMBL database, Tox21 dataset, and TOXNET dataset enables us to gather comprehensive information on bioactivity and toxicity, respectively. With the help of these data sources, we were able to study the ability of chemical compounds to inhibit the essential enzyme Acetylcholinesterase (AChE) as well as assess their safety profiles. This provided a good foundation for our subsequent investigation and modeling. The combination of many data sets from these reliable sources improves the precision and dependability of our study results and lays the framework for well-informed decision-making in the creation of prospective therapeutic approaches for Alzheimer's

Target Report Card

Name And Classification

ID: ChEMBL220

Type: SINGLE PROTEIN

Preferred Name: Acetylcholinesterase

Synonyms:

Synonyms: ---

organism: Homo sapiens

Species Group: No

Protein Target Classification:

1. Enzyme / Hydrolase

Figure 5.1: Upload File on the Frontend

Enhanced Molecular Generation

The ChEMBL database used in Prediction of bioactivity and toxicity has been employed in this component as well. The analysis of the dataset considered is as follows.

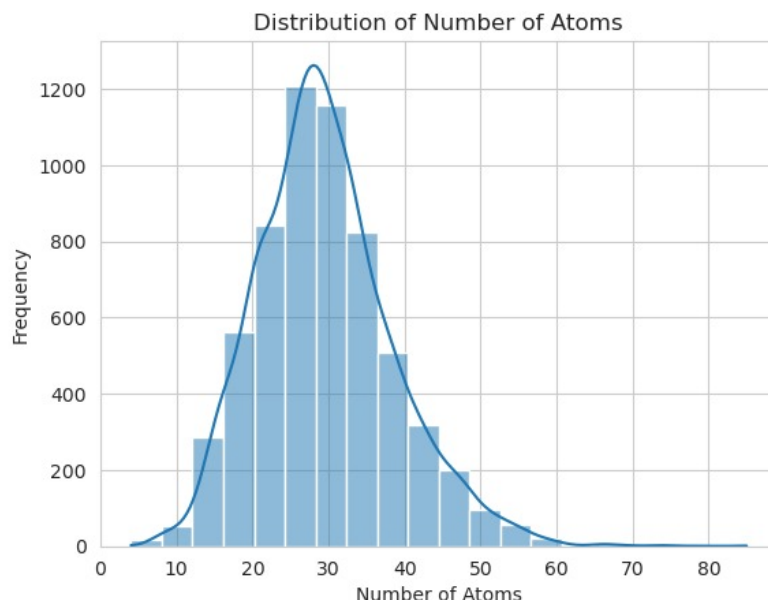


Figure 5.2: Graph showing the distribution of number of atoms

In examining the distribution of the number of atoms across the molecular structures, Fig.5.2. reveals an insightful pattern. The x-axis, denoting the number of atoms, begins from 0 and increments by 10, with a notable concentration observed in the range of 30 to 40 atoms, peaking at around 1200 instances. This distribution sheds light on the prevalent structural characteristics within the dataset.

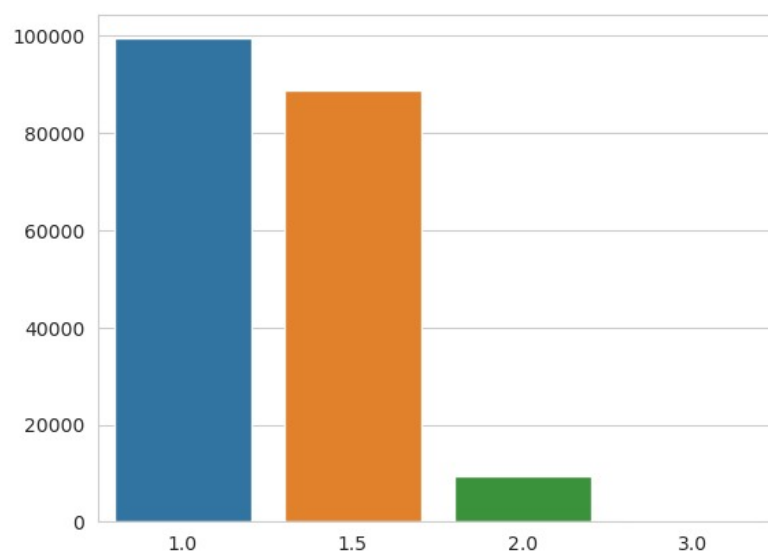


Figure 5.3: Graph showing the distribution of bond order

The analysis extends to the distribution of bond orders, providing a nuanced understanding of molecular connectivity, as illustrated in Fig.5.3. The x-axis, representing bond orders, is complemented by the y-axis depicting the count of elements. Notably, a bond order of 1.0 emerges with the highest frequency, nearing 100,000 occurrences, while a bond order of 1.5 on the x-axis exhibits the second-highest frequency, approaching 87,000 instances. This in-depth overview offers

perspectives on the prevailing bonding patterns within the molecular structures under consideration.

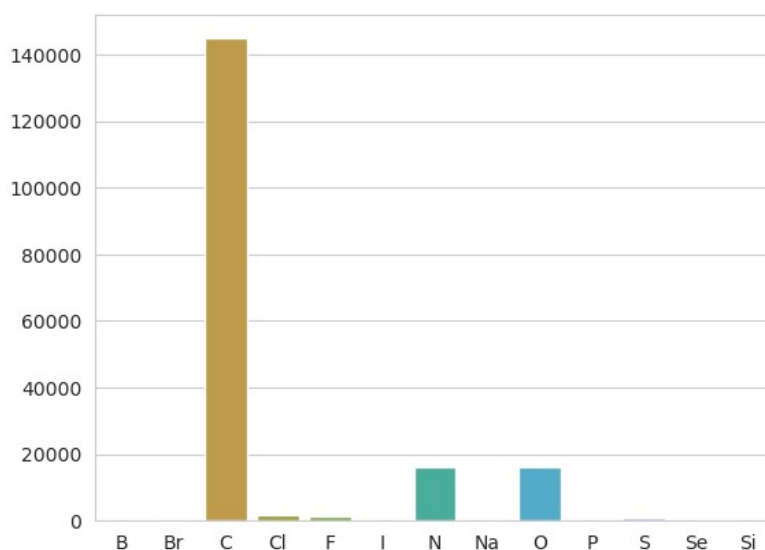


Figure 5.4: Graph showing the distribution of different types of elements

Additionally, an examination of the elemental composition within the initial dataset is undertaken. The x-axis encompasses elements such as B, Br, C, Cl, F, I, N, Na, O, P, S, Se, and Si, while the y-axis denotes the count of elements. Noteworthy is the prominence of carbon (C) in the dataset, with an approximate count of 140,000 instances, highlighting its prevalence among the chosen elements. These graphical representations deepen the understanding of the molecular characteristics inherent in the dataset, laying the foundation for subsequent AI-Driven analyses in drug discovery for Alzheimer's disease.

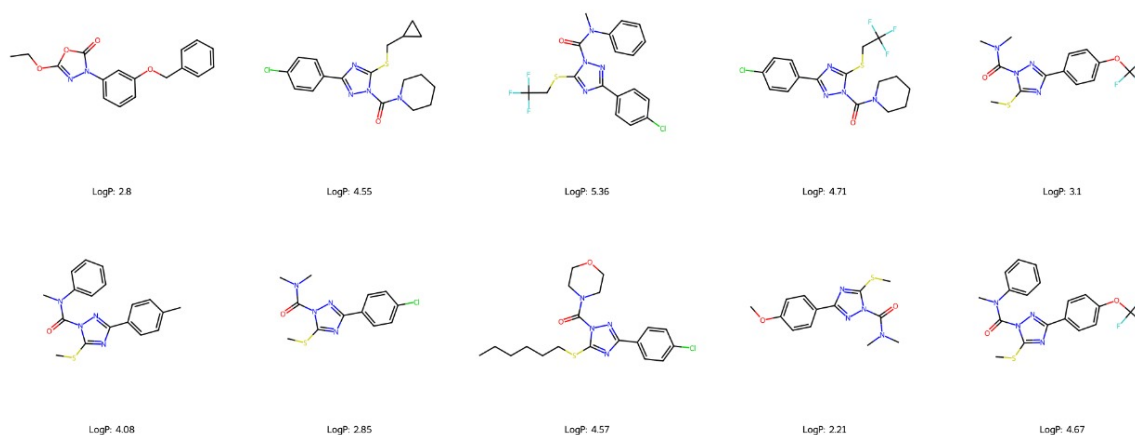


Figure 5.5: Illustrates the compounds within the initial, accompanied by their respective logP values.

Additionally, an examination of the elemental composition within the initial dataset is undertaken. The x-axis encompasses elements such as B, Br, C, Cl, F, I, N, Na, O, P, S, Se, and Si, while the y-axis denotes the count of elements. Noteworthy is the prominence of carbon (C) in

the dataset, with an approximate count of 140,000 instances, highlighting its prevalence among the chosen elements. These graphical representations deepen the understanding of the molecular characteristics inherent in the dataset, laying the foundation for subsequent AI-Driven analyses in drug discovery for Alzheimer's disease.

Prediction of Drug-Target Interactions

The dataset encompasses pairs of drugs and proteins, with a particular focus on their interactions with acetylcholinesterase. Alongside drug-protein pairs and their corresponding drug-target interaction (DTI) scores, this dataset also incorporates quantitative structure-activity relationship (QSAR) data. QSAR information provides valuable insights into the chemical properties of drugs and their relationship with bioactivity, enhancing our understanding of drug-target interactions. With approximately 160,000 samples, this dataset offers a rich source of information for developing predictive models that leverage both QSAR and DTI data. This comprehensive dataset is instrumental in advancing research on drug discovery and personalized medicine, particularly in the context of Alzheimer's disease treatment.

5.2 Environment setup(H/W and S/W)

5.2.1 Hardware Requirements:

- **High-performance Computing (HPC) Cluster:** A cluster of servers with multicore processors (Intel i5 or higher) and at least 8GB RAM (preferably DDR4) to handle computational workloads efficiently. More RAM and processing power are recommended for faster performance.
- **Graphics Processing Units (GPUs):** NVIDIA GTX 1660 or RTX 3060 GPUs with CUDA support for parallel processing and faster model training. Higher-end GPUs such as RTX 3080 or RTX 3090 are recommended for increased performance.
- **Solid State Drive (SSD):** At least 256GB SSD storage (preferably NVMe) for quick access and processing of large datasets. Larger SSD storage capacity is recommended to accommodate growing datasets and ensure optimal performance.

5.2.2 Software Requirements:

- **Python Programming Language:** Python 3.7 or later for AI development. Python 3.8 or 3.9 is recommended for compatibility with the latest libraries and frameworks.
- **Deep Learning Frameworks:** TensorFlow 2.5 or PyTorch 1.9 for building and training neural networks. The latest versions of these frameworks are recommended for improved features and performance.

- **Integrated Development Environment (IDE):** PyCharm Community Edition 2021.2 or Jupyter Notebook for writing and executing code. These IDEs offer advanced features for debugging, code navigation, and collaboration.
- **Operating System:** Linux (Ubuntu 20.04 LTS) or Windows 10 for compatibility with AI frameworks. Linux distributions such as Ubuntu or CentOS are preferred for better performance and stability.
- **Libraries:** Streamlit 0.87, NumPy 1.21, Pandas 1.3, SciKit-Learn 0.24 for data manipulation and visualization. Upgrading to the latest versions of these libraries is recommended to leverage new features and optimizations.

5.3 Performance Analysis

The Mann–Whitney U test is utilized to establish a threshold for the predicted PIC50 value, distinguishing between active ($\text{PIC50} > 6$) and inactive ($\text{PIC50} < 5$) compounds. This model effectively predicts PIC50 values, crucial for classifying compounds as bioactive or inactive. Compounds with PIC50 values above 6 are deemed bioactive, indicating significant inhibitory effects, while those below 5 are classified as inactive. This approach enables researchers to efficiently assess bioactivity and identify potential candidates for further investigation or drug development.

The Polypharmacology Browser is a computational tool used in drug discovery to explore the polypharmacological profiles of small molecules. It provides comprehensive information on a compound's interactions with multiple targets, offering insights into its potential therapeutic effects and side effects across various biological pathways. Tanimoto similarity, on the other hand, is a metric commonly used in cheminformatics to quantify the similarity between two chemical structures. It calculates the degree of overlap between their molecular fingerprints, providing a numerical measure of structural resemblance that aids in virtual screening and compound comparison tasks.

The efficacy of the models is assessed using a comprehensive set of metrics including accuracy, precision, F1 score, recall, and Matthews correlation coefficient (MCC). Accuracy measures the overall correctness of the model's predictions, while precision quantifies the proportion of true positive predictions among all positive predictions made by the model. F1 score balances precision and recall, providing a harmonic mean of these two metrics. Recall measures the proportion of true positive predictions among all actual positive instances in the dataset. Additionally, Matthews correlation coefficient evaluates the quality of binary classifications, considering true and false positives and negatives.

5.4 Results

Prediction of bioactivity and toxicity of molecules

The Random Forest algorithm emerged as the top performer among a range of models with an accuracy of 86%. Random Forest outperformed Decision Tree Regressor, Linear Regression, Ridge Regression, ExtraTreesRegressor, Bagging Regressor, XGB Regressor, AdaBoost Regressor, KNeighbors Regressor, and ExtraTreeRegressor in bioactivity prediction. For toxicity prediction, the Extra Tree Classifier exhibited superior performance compared to other classifiers, indicating its efficacy in discerning toxic and non-toxic compounds. These results highlight the robustness and reliability of Random Forest and Extra Tree Classifier models in predicting bioactivity and toxicity

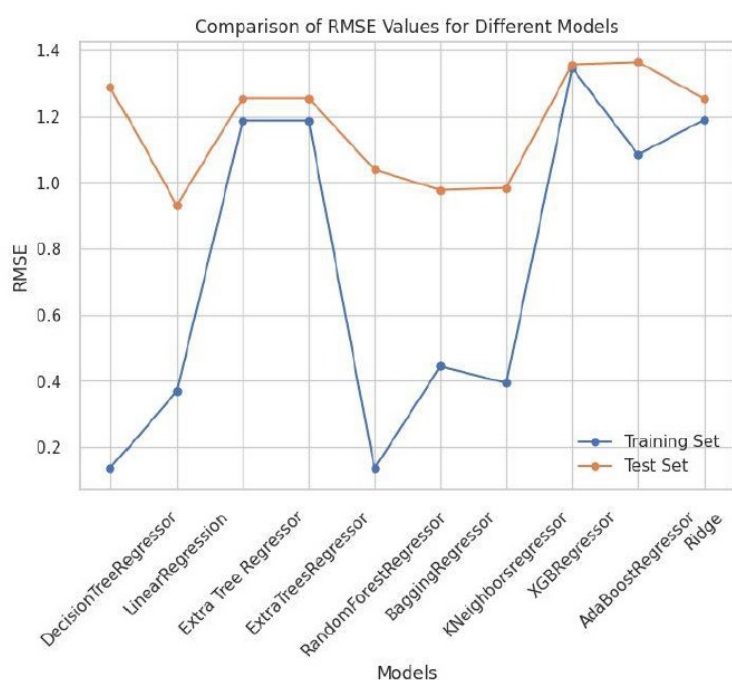


Figure 5.6: Graph representing the comparison of RMSE values for different regressor models

The residual graph of the Random Forest Regressor offers additional insight into the performance and predictive abilities. This enables a thorough evaluation of the precision and dependability of the machine learning models utilized in this study. The incorporation of these visuals improves the interpretation and comprehension of the results.

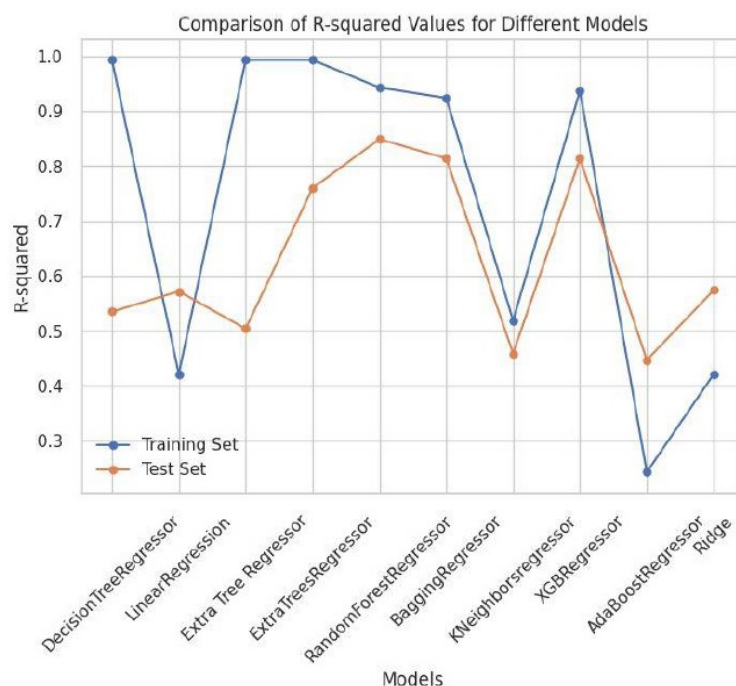


Figure 5.7: Graph representing the comparison of R-squared values for different regressor models

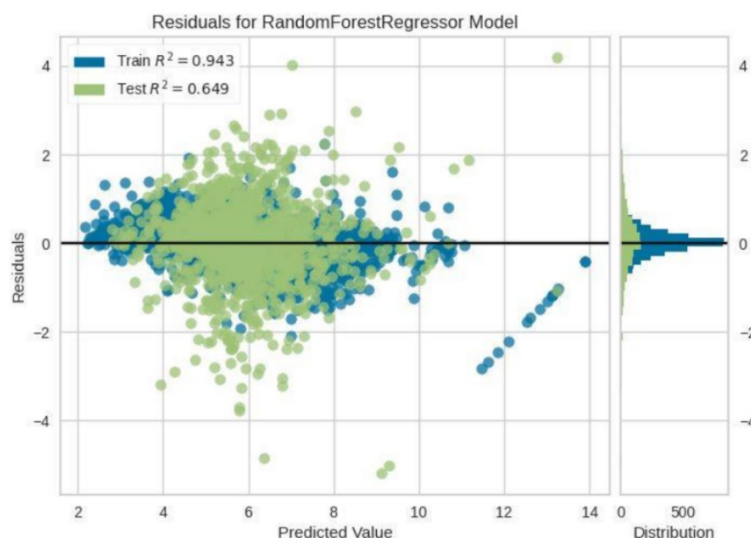


Figure 5.8: Residual Plot for Random Forrest Regressor Model

Enhanced Molecular Generation

The outcomes of the enhanced molecular generation component is evaluated. It is noted that the generator model exhibits strong performance, generating molecular structures with an impressive 84% validity in SMILES format. The successful transformation of molecular data into valid representations underscores the model's proficiency in capturing meaningful patterns. Additionally, the similarity in structure among the generated compounds was assessed, revealing an average Tanimoto similarity [8] of 0.70 when compared to molecules with known profiles in the Polypharmacology Browser 2 (PPB2) database [9]. This high similarity metric signifies the

potential relevance of the designed structures, aligning closely with established pharmacological entities within the PPB2 database. Complementing the generative process, the predictor model showcased robust performance metrics. The cross-Validated Root Mean Squared Error (RMSE) was notably low, measuring at 0.5734, indicating the model's accurate prediction of bioactivity levels. Additionally, the cross-Validated R2 score, a measure of the model's goodness of fit, was high at 0.9014. This signifies a robust ability to explain the variance in bioactivity predictions, highlighting the reliability and precision of our predictive model. The generator produced molecules biased toward high pIC50 values, subsequently validated by the predictor.

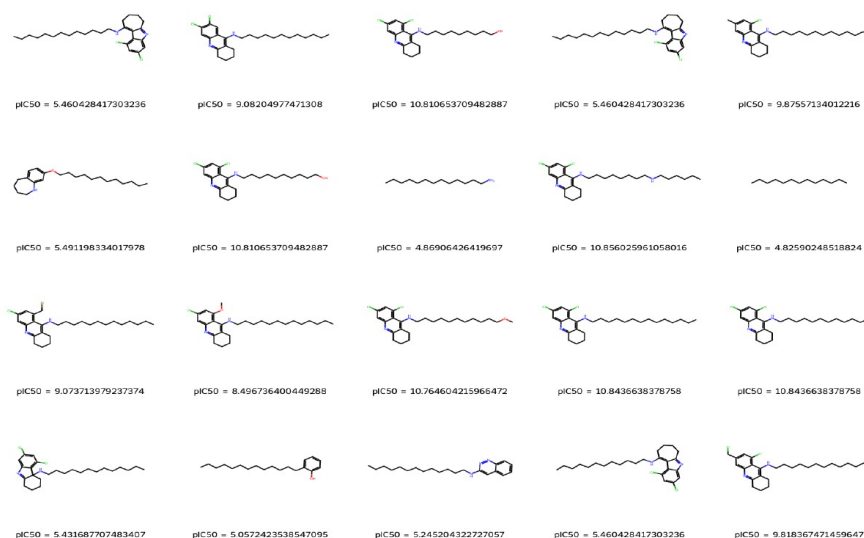


Figure 5.9: Molecules generated through the stacked GRU along with their respective pIC50 values.

Fig. 5.9. demonstrates molecules generated through the stacked GRU along with their respective pIC50 values.

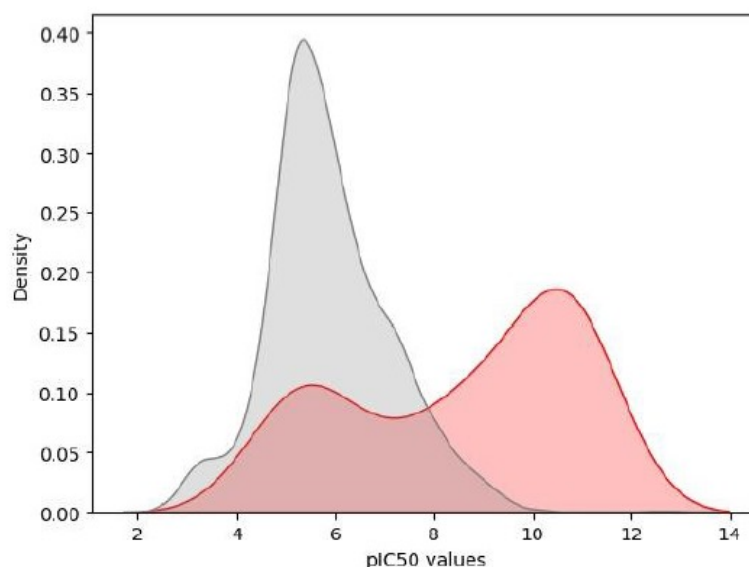


Figure 5.10: Graph showing the density of generated molecules in comparison to molecules in the initial dataset along with the associated pIC50 values

Fig. 5.10. illustrates a graph depicting the density of generated molecules in comparison to those in the initial dataset, along with their associated pIC50 values. The grey region represents molecules within the initial dataset, while the red region represents the designed molecules. Notably, a significant density of molecules in the grey region falls within the pIC50 value range of 4 to 6, whereas in the red region, a substantial density of designed molecules falls within the pIC50 value range of 10 to 12. This observation suggests that a greater number of designed molecules exhibit higher pIC50 values, implying that the generator model is proficient in producing molecules with elevated bioactivities.

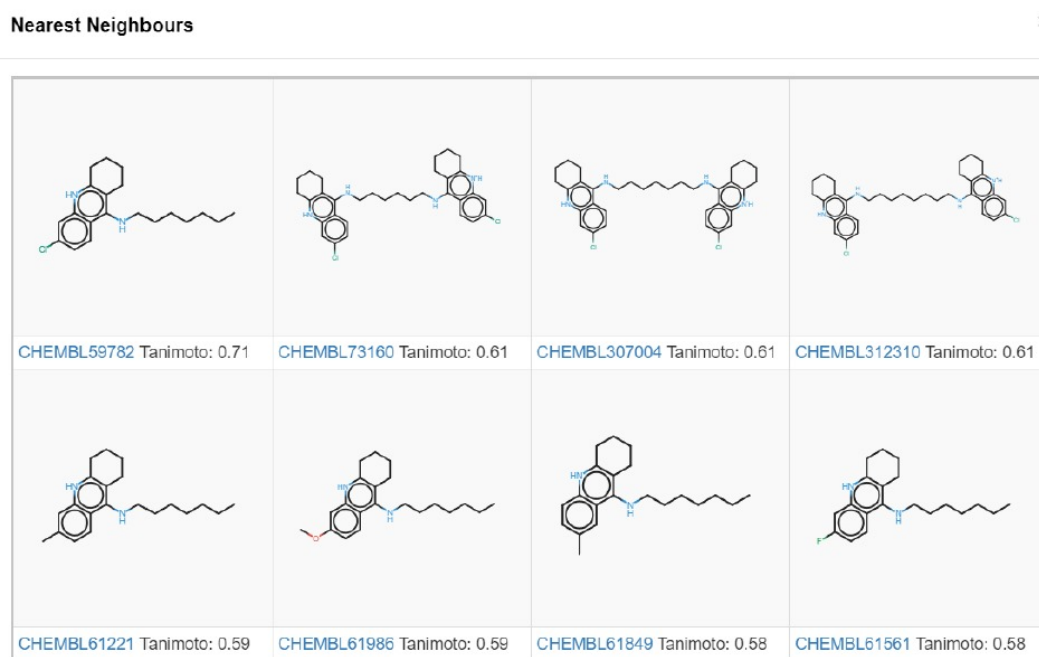


Figure 5.11: Molecules present in the Polypharmacology Browser 2 (PPB2) database which are structurally similar to the generated molecules

Fig. 5.11. displays the drug molecules that inhibit the breakdown of Acetylcholinesterase enzyme on the Polypharmacology Browser 2 (PPB2) database. These molecules are also structurally similar to the generated molecules. In this context, the maximum Tanimoto similarity observed is 0.70, and it corresponds to a compound with ChEMBL ID CHEMBL59782.

Prediction of Drug-Target Interactions

Several machine learning models underwent evaluation, with their training and testing performances calculated. Among them, Random Forest emerged as the top performer, demonstrating a remarkable success rate of 75%. The 3D structure of the Acetylcholinesterase protein was predicted by the utilization of AlphaFold in order to extract the protein sequence and structure embeddings.

Training Performance

	Set	Model	Accuracy	Recall	Precision	F1	Matthews Correlation Coe
0	Training Set (99705 DTIs)	Dummy Classifier	0.74 (0.71-0.76)	1.00 (1.00-1.00)	0.74 (0.71-0.76)	0.85 (0.83-0.87)	0 (0-0)
1	Training Set (99705 DTIs)	Logistic Regression	0.82 (0.79-0.84)	0.93 (0.91-0.95)	0.84 (0.81-0.86)	0.88 (0.86-0.90)	0.49 (0.42-0.55)
2	Training Set (99705 DTIs)	Linear Support Vector Classification	0.82 (0.79-0.84)	0.94 (0.92-0.95)	0.84 (0.81-0.86)	0.88 (0.87-0.90)	0.49 (0.42-0.55)
3	Training Set (99705 DTIs)	K-Nearest Neighbour Classifier	1.00 (1.00-1.00)	1.00 (1.00-1.00)	1.00 (1.00-1.00)	1.00 (1.00-1.00)	1.00 (1.00-1.00)
4	Training Set (99705 DTIs)	Decision Tree Classifier	1.00 (1.00-1.00)	1.00 (1.00-1.00)	1.00 (1.00-1.00)	1.00 (1.00-1.00)	1.00 (1.00-1.00)
5	Training Set (99705 DTIs)	Random Forest Classifier	1.00 (1.00-1.00)	1.00 (1.00-1.00)	1.00 (1.00-1.00)	1.00 (1.00-1.00)	1.00 (1.00-1.00)
6	Training Set (99705 DTIs)	Stochastic Gradient Descent Classifier	0.82 (0.79-0.84)	0.93 (0.91-0.95)	0.84 (0.81-0.86)	0.88 (0.87-0.90)	0.49 (0.43-0.55)

Figure 5.12: Training Performance of various classifiers employed for DTI

Testing Performance

	Set	Model	Accuracy	Recall	Precision	F1	Matthews Correlation Coefficient
0	Test Set (740 DTIs)	Dummy Classifier	0.68 (0.64-0.73)	1.00 (1.00-1.00)	0.68 (0.64-0.73)	0.81 (0.78-0.84)	0 (0-0)
1	Test Set (740 DTIs)	Logistic Regression	0.72 (0.68-0.76)	0.92 (0.89-0.95)	0.73 (0.69-0.78)	0.82 (0.79-0.84)	0.28 (0.18-0.37)
2	Test Set (740 DTIs)	Linear Support Vector Classification	0.72 (0.68-0.76)	0.85 (0.81-0.89)	0.77 (0.73-0.81)	0.80 (0.77-0.84)	0.32 (0.22-0.41)
3	Test Set (740 DTIs)	K-Nearest Neighbour Classifier	0.75 (0.71-0.79)	0.86 (0.82-0.90)	0.79 (0.75-0.83)	0.82 (0.79-0.85)	0.40 (0.31-0.48)
4	Test Set (740 DTIs)	Decision Tree Classifier	0.62 (0.58-0.66)	0.69 (0.64-0.74)	0.74 (0.69-0.79)	0.71 (0.67-0.75)	0.17 (0.07-0.26)
5	Test Set (740 DTIs)	Random Forest Classifier	0.76 (0.72-0.80)	0.92 (0.89-0.95)	0.77 (0.73-0.81)	0.84 (0.81-0.87)	0.41 (0.32-0.50)
6	Test Set (740 DTIs)	Stochastic Gradient Descent Classifier	0.73 (0.69-0.77)	0.92 (0.88-0.94)	0.75 (0.70-0.79)	0.82 (0.79-0.85)	0.31 (0.22-0.40)

Figure 5.13: Testing Performance of various classifiers employed for DTI

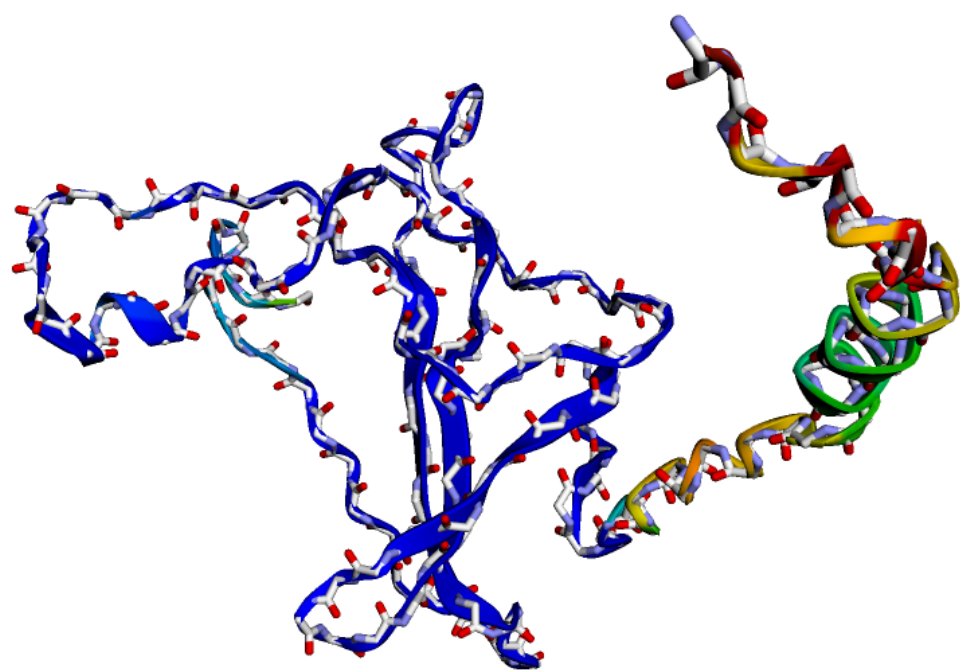


Figure 5.14: 3D structure of Acetylcholinesterase

Chapter VI

Conclusion and Future scope

The project underscores the transformative role of AI in revolutionizing drug discovery. Advanced machine learning techniques have facilitated the identification of potential drug candidates. Predictive modeling has played a pivotal role in bioactivity and toxicity prediction, enabling the prioritization of molecules with optimal pharmacological profiles for further investigation. Additionally, AI-driven approaches to enhanced molecular generation have shown promise in generating novel compounds with improved therapeutic capabilities, while classification algorithms for drug-target interaction prediction have enhanced understanding of molecular interactions, paving the way for the development of targeted therapeutics.

Furthermore, the project emphasizes the importance of accurate bioactivity and toxicity prediction in drug discovery. By assessing the bioactivity and toxicity profiles of candidate molecules using Random Forest predictor and Extra Tree Classifier respectively, potential risks associated with adverse effects have been mitigated, and the selection of promising drug candidates has been optimized.

Additionally, significant progress has been made in enhancing molecular generation for drug design purposes. By employing advanced generative models such as stacked Gated Recurrent Unit (GRU) architecture and reinforcement learning, diverse molecular structures optimized for targeting specific diseases have been generated. This innovative approach holds immense promise for accelerating the discovery of novel drug candidates and expanding the therapeutic arsenal available for treating complex diseases.

Lastly, advancements have been achieved in drug-target interaction prediction through the integration of advanced computational methods. Incorporating protein structure embeddings obtained from state-of-the-art techniques such as AlphaFold has enhanced the accuracy and robustness of prediction models, enabling more precise identification of potential drug targets. The findings underscore the importance of leveraging multidisciplinary approaches and cutting-edge technologies to unlock new opportunities

Going forward, utilizing scientific simulations is integral for predicting the behavior of drug compounds within biological systems, aiding researchers in comprehending how these compounds interact with target proteins and anticipating both their efficacy and potential side effects. With

the integration of AI into these simulations, the vast quantities of data involved can be efficiently analyzed, enabling the identification of patterns and the formulation of accurate predictions. Moreover, AI's application in retrosynthesis planning, a pivotal process in organic chemistry, particularly in drug development, offers significant advancements. Retrosynthesis planning entails deconstructing complex molecules, such as drug compounds, into simpler precursor molecules. Here, AI algorithms emerge as invaluable tools, automating and optimizing the retrosynthesis planning process. By scrutinizing extensive databases of chemical reactions and molecular structures, AI can intelligently suggest viable pathways for synthesizing target molecules from readily available starting materials, expediting the drug discovery process and enhancing efficiency in organic synthesis endeavors.

Chapter VII

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Chapter VIII

Bibliography

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The source code and the application of PharmAIcist can be viewed here:

github.com/ShreyasKuntal/PharmAIcist

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